

Small Pictures of Difference Maps

Basic Building Blocks, in Algebra, Geometry and Combinatorics

Abstract

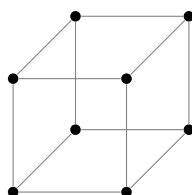
This note collects the *simplest* pieces behind a Lean 4 library on difference maps over groups of exponent two. Each piece is shown three ways: a short formula, a small picture, and the matching Lean snippet. Nothing here needs deep theory — every block already lives in Mathlib.

1 The carrier: a cube

The carrier is a finite abelian group where every element is its own inverse:

$$a + a = 0 \quad \text{for all } a.$$

Such a group is just a vector space over the two element field. A group of size 2^n is the *Boolean cube* $\{0,1\}^n$, with bitwise XOR as addition.



The group of size 2^3 : eight points, edges = flip one bit.

In the library the carrier is axiomatised (no named algebra is assumed):

```
ax_prod_comm : a b, prod_ a b = prod_ b a      -- abelian
ax_prod_unit : a, prod_ unit_ a = a            -- has a zero
ax_prod_inv  : a, prod_ (inv_ a) a = unit_      -- has inverses
ax_char2     : a, prod_ a a = unit_            -- exponent two
```

2 Freshman's dream: a commuting square

In characteristic two, squaring is *additive*:

$$(a + b)^2 = a^2 + b^2.$$

So the squaring map $\varphi(x) = x^2$ (the Frobenius map) makes a square commute:

$$\begin{array}{ccc} G \times G & \xrightarrow{+} & G \\ \varphi \times \varphi \downarrow & & \downarrow \varphi \\ G \times G & \xrightarrow{+} & G \end{array}$$

This is already in Mathlib:

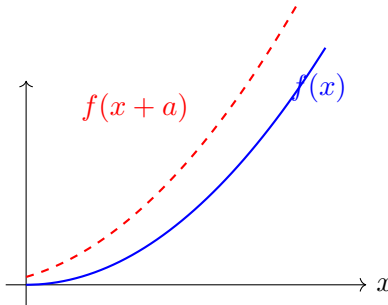
```
-- Mathlib: in characteristic p,
add_pow_char : (a + b) ^ p = a ^ p + b ^ p
```

3 The difference map

Fix a direction a . The difference of a map f in direction a is

$$D_a f(x) = f(x + a) - f(x).$$

Geometrically: slide the graph of f left by a , then subtract.



The dashed curve is $f(x + a)$; the difference is the gap.

The library writes this for both a concrete field and an abstract carrier:

```
def D (f : F → F) (a x : F) : F := f (x + a) - f x

def tDiff (d : →) (a x : S.obj) : S.obj :=
  S.prod_ (S.endo d (S.prod_ x a)) (S.inv_ (S.endo d x))
```

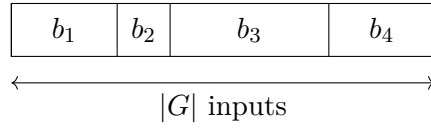
4 Fibers and width: a partition

Fix a direction a and a value b . The *fiber* is the set of inputs that the difference sends to b , and the *width* is its size:

$$\text{fib}(a, b) = \{x : D_a f(x) = b\}, \quad \text{wid}(a, b) = |\text{fib}(a, b)|.$$

As b ranges over the group, the fibers tile the whole carrier:

$$\sum_b \text{wid}(a, b) = |G|.$$



Each block is one fiber; the block widths add up to $|G|$.

```
def fib (d : ) (a b : S.obj) : Finset S.obj :=
  univ.filter (fun x => tDiff S d a x = b)

def wid (d : ) (a b : S.obj) : := (fib S d a b).card

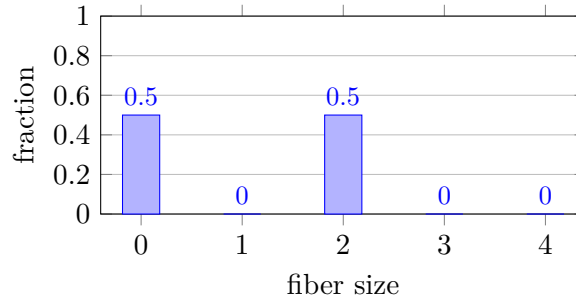
theorem wid_partition (d : ) (a : S.obj) :
  b, wid S d a b = Fintype.card S.obj
```

5 APN: width at most two

A map is *almost perfect nonlinear* (APN) when every fiber in every nonzero direction has size 0 or 2:

$$\text{wid}(a, b) \in \{0, 2\} \quad (a \neq 0).$$

So the *width spectrum* is very flat — only two bars:



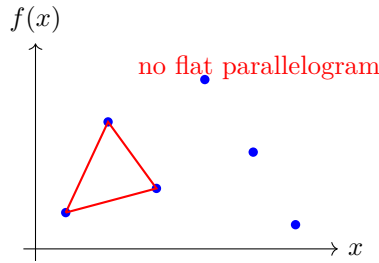
An APN map: half the values are missed, half are hit exactly twice.

```
def IsUniform ( : ) (f : F → F) : Prop :=
  a, a ≠ 0 → ∃! b, N f a b = 0 → N f a b =

def IsAPN (f : F → F) : Prop := IsUniform 2 f
```

6 The graph as a Sidon set

Read the same condition in additive combinatorics. The *graph* $\{(x, f(x))\}$ is a *Sidon set*: it contains no nontrivial parallelogram, i.e. four graph points adding to zero must already be paired up.



```
def IsSidonGraph (f : F → F) : Prop :=
  p p p p : F × F,
  isInGraph f p isInGraph f p
  isInGraph f p isInGraph f p
  p + p + p + p = 0
  (p = p p = p) (p = p p = p) (p = p p = p)
```

7 Maps between systems: a category

Finally, systems and the maps that respect their structure form a category. A morphism keeps addition and commutes with the endomorphism family:

$$\begin{array}{ccc} S & \xrightarrow{f} & T \\ \text{endo}_d \downarrow & & \downarrow \text{endo}_d \\ S & \xrightarrow{f} & T \end{array}$$

```
structure Mor (S T : EndoSys) where
  f : S.obj → T.obj
```

```

pres : a b, f (S.prod_ a b) = T.prod_ (f a) (f b)
nat   : d x, f (S.endo d x) = T.endo d (f x)

```

Equivalence classes of such maps split into a finer level (affine, EA) sitting inside a coarser one (CCZ):

$$\text{EA} \subsetneq \text{CCZ}.$$

Every block above is elementary; together they are the alphabet of the library.